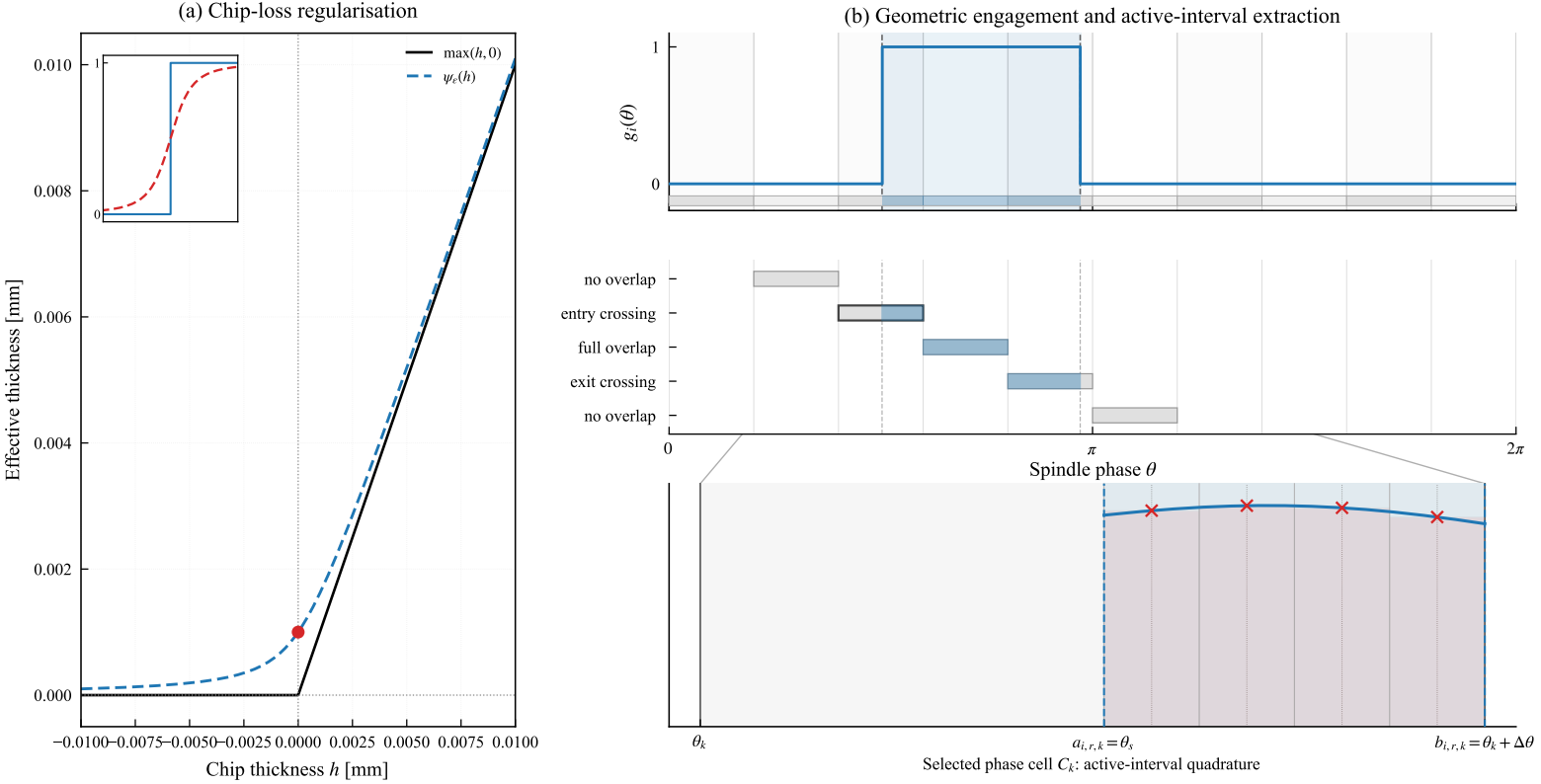
(a) Chip-loss regularisation
max(h, 0)
ψε(h)
Effective thickness [mm]
Chip thickness h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
(b) Geometric engagement and active-interval extraction
gi(θ)
no overlap
entry crossing
full overlap
exit crossing
no overlap
0
π
2π
Spindle phase θ
θk
ai,r,k = θs
bi,r,k = θk + Δθ
Selected phase cell Ck: active-interval quadrature